

SERIAL COMMUNICATIONS WITH AVR MICROCONTROLLER

■ ARUN DAYAL UDAI

Serial communication is a low-level communication between two or more devices. Normally, one of the devices is a computer, while the other device can be a modem, printer, another computer, or a scientific instrument such as an oscilloscope or a function generator. It may even be a device to device communication like that between a microcontroller and a sensor, LCD, storage memory or a different microcontroller.



In serial communication, you can send and receive data serially, one bit at a time. In a microcontroller, it can be realised in two modes of serial communication—synchronous and asynchronous.

The article covers the concept in three broad sections—synchronous serial communication, asynchronous

serial communication and serial port interfacing using PC through GUI developed in Visual Basic.Net

Synchronous serial communication

Serial peripheral interface (SPI) is a synchronous serial data communication that includes a three-wire transmission mode, with master out slave in (MOSI), master in slave out (MISO) and synchronising clock (SCK) pins. Ground (GND) acts as a common reference and slave select ($\overline{SS}$) selects the slave which is to be driven by the master, by setting it low. $\overline{SS}$ pin may be permanently set to low if only one slave is being used. Many slaves are connected in parallel and the master selects one out of many by setting the slave's $\overline{SS}$ pin to low. MOSI and MISO forms a cyclic shift register which is moved bit-wise with the synchronising

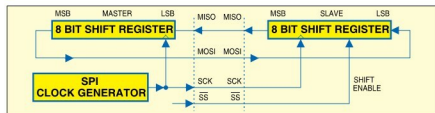


Fig. 1: SPI master-slave interconnection

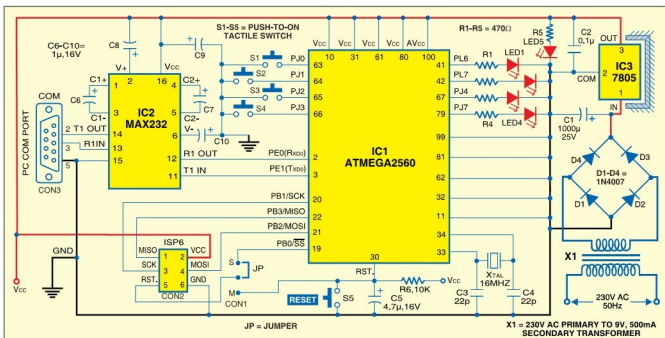


Fig. 2: Circuit of synchronous serial communication with AVR microcontroller